

Science Olympiad
MIT Invitational
January 24, 2026
Astronomy C



Directions:

- Each team will be given **50 minutes** to complete the exam.
- There are **74 questions** split into two sections: **§A** (Multiple Choice) and **§B** (Free Response).
- **Do not write on the exam or image sheet.** Only write on your answer sheet.
- For calculation questions, **work will be graded.** Please show all your work, and box your final answer.
- The use of AI tools (e.g. ChatGPT) are expressly forbidden.
- Tiebreakers, in order: §B-VI, §B-V, §B-IV, §B-II, §B-III, §B-I, §A.
- Good luck! And may the stars align for you.

Written by: The Astronomy A-Team

April Cheng, aqcheng@princeton.edu | **Sahil Pontula**, spontula@mit.edu | **Brian Sun**,
briansun@mit.edu | **Rio Sessions**, rio.sessions@student.nmt.edu

Section A: Multiple Choice

This section consists of 30 multiple choice questions about general astronomy concepts. Each question is worth 2 points, for a total of 60 points.

- Which spectral type shows the strongest hydrogen Balmer absorption lines?
 - O
 - B
 - A
 - F
 - M
- Which MK luminosity class corresponds to supergiant stars?
 - I
 - III
 - V
 - VII
 - IX
- What is the dominant energy source for a protostar *before* hydrogen fusion ignites?
 - Hydrogen fusion
 - Gravitational contraction
 - Helium shell burning
 - Electron degeneracy pressure
 - Neutrino cooling
- A $0.6 M_{\odot}$ star is currently on the main sequence. When it exhausts its core hydrogen, which evolutionary path will it follow?
 - Main sequence $\rightarrow$ Planetary nebula and white dwarf
 - Main sequence $\rightarrow$ Red giant $\rightarrow$ Planetary nebula and white dwarf
 - Main sequence $\rightarrow$ Red giant $\rightarrow$ Type II supernova $\rightarrow$ White dwarf
 - Main sequence $\rightarrow$ Red supergiant $\rightarrow$ Type II supernova $\rightarrow$ Neutron star
 - Main sequence $\rightarrow$ Red or blue supergiant $\rightarrow$ Type Ib/c supernova $\rightarrow$ Nothing
- T Tauri stars are best described as:
 - Young, accreting, irregularly variable stars
 - Red supergiants
 - Degenerate helium-burning stars
 - Carbon-rich AGB stars
 - White dwarf precursors on the cooling curve
- Which statement best describes the Hayashi track?
 - The path a star takes as it fuses hydrogen into helium
 - The nearly vertical track on the H-R diagram representing fully convective pre-main sequence contraction
 - The horizontal evolution of a star during helium flash
 - The rapid expansion phase during planetary nebula ejection
 - The cooling track of white dwarfs
- Mira variables typically have pulsation periods of:
 - < 1 day
 - 1–10 days
 - 100–500 days
 - 10–20 years
 - Irregular, non-periodic cycles
- Mechanistically, type Ia supernovae are most similar to
 - Tidal disruption events
 - Type Ib supernovae
 - Type IIa supernovae
 - X-ray bursts from low-mass x-ray binaries
 - The helium flash of a red giant star

9. What parameter primarily determines a star's position on the main sequence and subsequent evolution?
- A. Radius
 - B. Chemical composition
 - C. Mass
 - D. Rotation rate
 - E. Surface gravity
10. Normal stars are supported against gravity by
- A. Thermal pressure
 - B. Radiation pressure
 - C. Neutron degeneracy pressure
 - D. Electron degeneracy pressure
 - E. Both A and B
11. Which statement about white dwarfs is TRUE?
- A. More massive white dwarfs have larger radii
 - B. White dwarfs are supported by electron degeneracy pressure
 - C. White dwarfs are actively fusing helium in their cores
 - D. All white dwarfs eventually explode as Type II supernovae
 - E. White dwarfs are hotter than main sequence O-type stars
12. Suppose an observation of a pulsar over time reveals that its rotational period is slowly increasing. Where is its angular momentum and rotational energy likely going to?
- A. A companion star
 - B. An accretion disk
 - C. The magnetic field of the pulsar
 - D. The plasma of the surrounding interstellar medium
 - E. Nowhere; the pulsar has simply expanded in radius
13. What is the mechanism by which this happens?
- A. Roche lobe overflow
 - B. Accretion
 - C. Magnetic braking
 - D. A "starquake"
 - E. None of the above
14. The distance in parsecs to a star with parallax p milliarcseconds is:
- A. $d = p$
 - B. $d = 1/p$
 - C. $d = 1/p^2$
 - D. $d = 0.001/p$
 - E. $d = 1000/p$
15. In a spectroscopic binary, the star with the larger radial-velocity amplitude is:
- A. Less massive
 - B. More massive
 - C. Hotter
 - D. More luminous
 - E. More evolved
16. RR Lyrae stars are commonly found in:
- A. Young open clusters
 - B. Old globular clusters
 - C. Spiral arms
 - D. Star-forming regions
 - E. Galactic halo gas clouds
17. The primary energy source of horizontal branch stars is
- A. Hydrogen fusion in the core
 - B. Helium fusion in the core
 - C. Hydrogen fusion outside the core
 - D. Gravitational contraction
 - E. Deuterium fusion

18. Pulsars are most commonly observed in which wavelength?

- A. Visible
- B. Infrared
- C. Radio
- D. UV
- E. Gamma-ray

19. The instability strip on the HR diagram contains which of the following?

- A. Cepheid variables
- B. Mira variables
- C. T Tauri stars
- D. Asymptotic Giant Branch stars
- E. Wolf-Rayet stars

20. What is the common mechanism driving the variable stars on the instability strip?

- A. Opacity changes driven by the partial ionization of hydrogen
- B. Opacity changes driven by helium ionization levels
- C. Star spots rotating in and out of view
- D. A pulsating stellar envelope due to repeated helium shell ignitions
- E. Binary interactions in a highly eccentric orbit

21. What feature explains the double-lobed radio structure of some massive stars?

- A. Thermonuclear flashes
- B. Stellar rotation
- C. Collimated outflows or jets
- D. Binary eclipses
- E. Planetary nebula shells

22. A neutron star in an X-ray binary is fed material from:

- A. Stellar winds only
- B. Dusty envelopes
- C. Accretion disks

D. Radiation pressure

E. Free-fall collapse

For questions 23-27, refer to the cover image.

23. One can see in this image large numbers of young, blue stars. Such a nebula could eventually evolve into a(n)

- A. Open cluster
- B. Globular cluster
- C. Nuclear star cluster
- D. Dwarf galaxy
- E. None of the above

24. The red, diffuse emission you see in the cover image is a signature of star formation. This is because it implies

- A. a large population of red giants, indicating a quickly evolving population
- B. strong dust extinction, which preferentially absorbs blue light and reddens the nebula
- C. thermal emission from cool molecular gas heated during gravitational collapse
- D. the existence of young, hot O/B stars that produce ionizing radiation
- E. synchrotron radiation from relativistic electrons accelerated by recent supernovae

25. This emission line is caused by an electron transitioning from which energy level to which energy level of which element?

- A. Hydrogen, $3 \rightarrow 2$
- B. Hydrogen, $4 \rightarrow 2$
- C. Helium, $3p \rightarrow 2s$
- D. Helium, $3d \rightarrow 2p$
- E. Helium, $2p \rightarrow 2s$

26. Why is optical emission dominated by this transition over other transitions?

- A. Higher energy initial states are less common
- B. Lower energy initial states are less common
- C. Transitions with lower energy final states are not in the optical band
- D. Both A and C
- E. Both B and C

27. Finally, identify this DSO.

- A. Orion Molecular Cloud Complex
- B. Sharpless 29
- C. Ophiion Star Family
- D. Helix Nebula
- E. Cassiopeia A

The sun has been slowly increasing in temperature and luminosity since its birth around 5 billion years ago due to the depletion of core hydrogen. For questions 28-30, you may take the current temperature of the sun to be 5772 K.

28. If one were to take the spectrum of sun at zero-age main sequence (ZAMS), one would find the spectral density to peak around 516 nm. How much hotter is the sun today than at ZAMS?

- A. 1%
- B. 3%
- C. 7%
- D. 14%
- E. 25%

29. The sun today is around 50% brighter than it was at ZAMS. Using your answer to the previous question, what was the radius of the sun at ZAMS? Give your answer in solar radii.

- A. $0.85 R_{\odot}$
- B. $0.95 R_{\odot}$
- C. $1.05 R_{\odot}$
- D. $1.15 R_{\odot}$
- E. $1.25 R_{\odot}$

The equilibrium temperature of the Earth—the temperature of the Earth if incoming solar radiation and outgoing blackbody radiation are equal, ignoring atmospheric effects—is 255 K, or -18° C. (Of course, the Earth is warmer than that due to the greenhouse effect.)

30. How bright would the sun need to be for Earth's water to boil (at atmospheric pressure), even without any greenhouse effect?

- A. $1.5 L_{\odot}$
- B. $3 L_{\odot}$
- C. $5 L_{\odot}$
- D. $7 L_{\odot}$
- E. $10 L_{\odot}$

Given the current rate at which the sun is getting brighter, this would seem to imply that we have at least until the end of the sun's main sequence before Earth's oceans boil away. Unfortunately, this is projected to occur much sooner (as soon as 1-2 Gyrs) from a runaway greenhouse effect.

B-II: WDJ181058.67+311940.94 [20 pts]

36. [3 pts] First of all, this is quite a long name for a DSO! What do all of those letters and numbers mean?
37. [3 pts] This DSO is a binary system, comprised of a primary star of mass $M_1 = 0.834 \pm 0.039 M_\odot$ and a secondary star of mass $M_2 = 0.721 \pm 0.020 M_\odot$. Compute the total mass of the system and its uncertainty. Is it possible for this system to go supernova? Why or why not?

Fig. 2 depicts a spectrum from this system, whose two components have effective temperatures around 20,000 K.

38. [2 pts] What does the unit “mJy” represent on the y axis? Don’t simply expand the abbreviation, but instead explain what this unit refers to.
39. [1 pt] What spectral line is depicted?
40. [2 pts] What causes the slight bump in the spectrum around 1400 angstroms?
41. [2 pts] Would you expect this bump to persist if the effective temperature was instead 30,000 K? Why or why not?
42. [4 pts] Much like a black hole or neutron star binary, this system will emit gravitational waves and eventually merge. This timescale can be written as

$$T_{\text{merge}} = \alpha f(a_0, M_1, M_2), \quad (1)$$

where $\alpha = \frac{5}{256}$ is a dimensionless constant and $f(a_0, M_1, M_2)$ is a function of the initial separation between the two stars, their masses, and fundamental constants. Use dimensional analysis and any other reasonable physical arguments to completely identify the function f .

43. [3 pts] Use your result to estimate the merger time for WDJ181058.67+311940.94, given $a_0 = 0.016$ AU. Is your result larger or smaller than the current age of the universe?

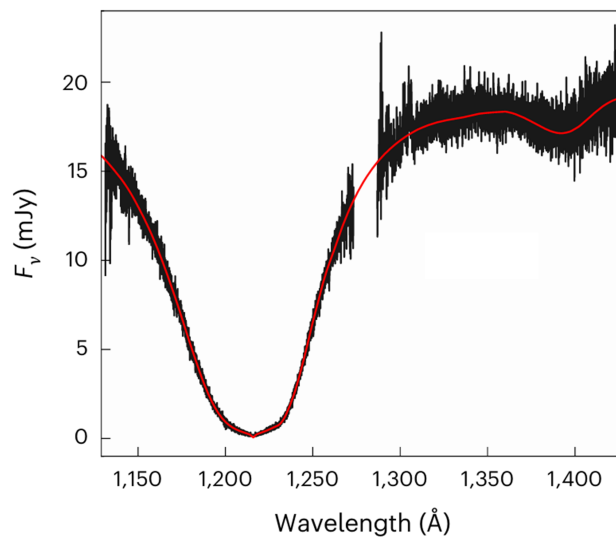


Figure 2: Figure for B-II.

B-III: Broken bones [17 pts]

44. [1 pt] One of your DSOs is eloquently named “The Bone.” In which galaxy is this DSO found?

Fig. 3a depicts this DSO with data from multiple parts of the electromagnetic spectrum.

45. [2 pts] Which part of the electromagnetic spectrum corresponds to the central filament (a.k.a. “the bone” or “the snake”) that extends from the top to the bottom of the image? Which telescope(s) collected data corresponding to this feature?

46. [2 pts] Describe the physical mechanism that gives rise to the filamentary structure.

47. [3 pts] Which part of the electromagnetic spectrum corresponds to the bright blue object to the right of the filament? Which telescope(s) collected data corresponding to this feature? Why is this feature blue?

48. [2 pts] If you look carefully at the “fracture” in the filament near the center of the image, a bright spot appears just to the right of it. What is the source of this bright spot?

Fig. 3b depicts the radio spectrum for the bright spot. We define the spectral index α through $S_\nu \propto \nu^\alpha$, where S_ν is the flux density at frequency ν .

49. [2 pts] What is the value of α for a blackbody at radio frequencies?

50. [2 pts] Use the graph in Fig. 3b to calculate the value of α for this object.

51. [3 pts] Values of α between roughly -0.1 and 2 correspond to thermal emission. Does the object show thermal or non-thermal emission? If non-thermal, what is the physical process most likely to give rise to this value of α ?

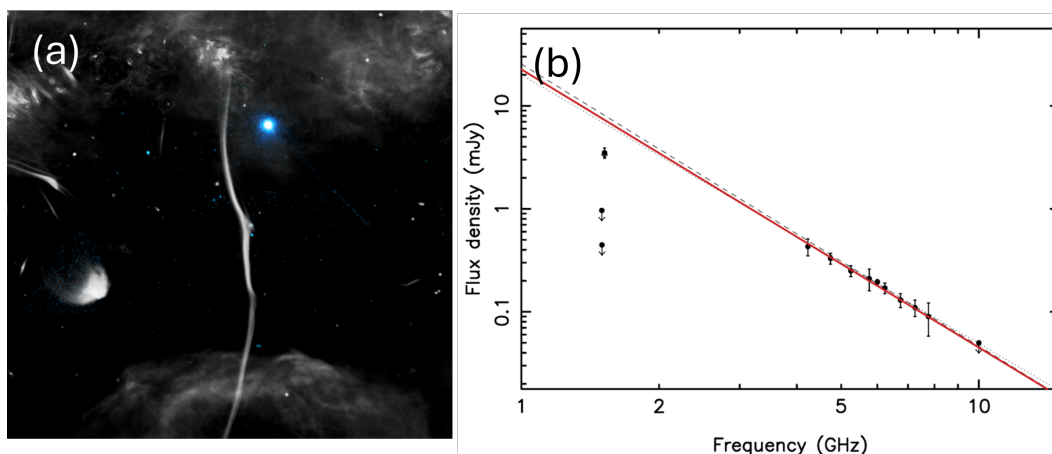


Figure 3: Figures for B-III.

B-IV: Modeling Mira's Pulsations [24 pts]

Parts (a) and (b) can be solved separately.

(a) Scaling Laws

52. [3 pts] Let us model the timescale of Mira's radial pulsations with only G , M , and R . That is, let $P \propto G^a M^b R^c$. Use dimensional analysis to find exponents a, b, c .
53. [2 pts] Assume for part (a) only that a Mira variable oscillates vertically on a H-R diagram. Find α, β for a new model $P \propto L^\alpha M^\beta$.
54. [4 pts] Scientists have determined this empirical period-luminosity model for K-band magnitudes:

$$M_K = -3.52(\log P - 2.38) - 7.25.$$

Ignoring bolometric correction, compare your scaling law from the previous question to this formula. What does this say about the Mira's pulsation mode?

(b) A Time Dependent Model

55. [3 pts] Let Mira's radius vary as $R(t) = R_0 + \Delta R(t)$, $|\Delta R(t)| \ll R_0$, where ΔR is governed by a simple harmonic oscillator of frequency $\omega = 2\pi/P$:

$$\frac{d^2(\Delta R)}{dt^2} + \omega^2 \Delta R = 0.$$

Express $\frac{R(t)}{R_0}$ in terms of normalized radius amplitude $\epsilon_R = \frac{\Delta R}{R_0}$ and phase ϕ .

56. [4 pts] To the first order approximation, assume that we have a surface temperature-radius relation

$$\frac{\Delta T}{T} = -\gamma \frac{\Delta R}{R}.$$

Approximate the fractional luminosity amplitude ϵ_L in terms of ϵ_R and γ .

57. [5 pts] Using your result from the last part, derive an expression for the magnitude variation $\Delta M_K(t) \equiv M_K(t) - M_{K,0}$ in terms of $\epsilon_R, \gamma, \omega, \phi$, and constants. You may make first order approximations.

Hint: For $|x| \ll 1$, $\log_{10}(1+x) \approx x/\ln 10$.

58. [3 pts] Using your expression for $\Delta M_K(t)$, determine whether the light curve is in phase or out of phase with the radius curve $R(t)$ as a function of γ .

B-V: ZTF J203349.8+322901.1 [26 pts]

We explore two theories for the different hemisphere compositions of Janus.

(a) Transitioning Layers

59. [1 pt] Compute the surface gravity g of Janus ($M = 1.2M_{\odot}$, $R = 3400$ km, $T_{\text{eff}} = 3.0 \times 10^4$ K)
60. [7 pts] From hydrostatic equilibrium $dP/dz = -\rho g$ and the equation for an ideal gas, derive the scale height H of the hydrogen ($m \approx m_p$) and helium ($m \approx 4m_p$) atmospheres. Scale height is defined as the unique parameter for pressure/density profiles:

$$P(z) = P_0 \exp\left(-\frac{z}{H}\right)$$

61. [2 pts] Argue why, in the absence of strong mixing, a multicomponent atmosphere would tend to become stratified with lighter elements (e.g. H) over heavier elements (e.g. He).

(b) Magnetic Asymmetry

62. [3 pts] Qualitatively, how would an asymmetric magnetic field cause a difference in composition?
63. [4 pts] Define optical depth with $d\tau = \kappa \rho dz$, where Rosseland opacity $\kappa = 0.02 \cdot (1 + X)$ m²/kg is a function of hydrogen mass fraction X . Estimate the photospheric gas pressure P_{gas} by integrating $dP/d\tau$ from $\tau = 0 \rightarrow 2/3$.
64. [5 pts] Let's model one hemisphere with flux density $B = 0$ and the other with $B = \Delta B$. Using the fact that both hemispheres must have equal $P_{\text{total}} = P_{\text{gas}} + P_B$ in equilibrium, calculate an approximation for ΔB .
- Hint:* P_B is equivalent to magnetic energy density ($[\frac{\text{N}}{\text{m}^2}] = [\frac{\text{J}}{\text{m}^3}]$).
65. [4 pts] For hydrogen lines, Zeeman splitting $\Delta\lambda$ is often modeled as

$$\Delta\lambda \sim 4.7 \times 10^{-13} \lambda_0^2 B$$

with λ_0 and $\Delta\lambda$ in Å and B in Gauss. With the knowledge that we cannot detect Zeeman splitting for Janus' H- α line, what is an upper bound to ΔB ? Assume observations are taken with spectral resolution $\Delta\lambda_{\text{res}} = 10.0$ Å.

B-VI: Sputtering X-Ray Binary [22 pts]

In this question, we use JS9 to analyze x-ray observations of an x-ray binary, with a neutron star that undergoes short bursts of accretion from a companion object.

To open the file, go to chandra.si.edu/js9. In the JS9 window, select **File > Open Remote**. Enter bit.ly/js9-mit26-xrb. Select **Open**, and wait for the file to load in.

FITS Header

66. [2 pts] From the FITS Header file, determine the following.

- (a) The telescope that took this image.
- (b) The name of the object observed here.

Light Curve

To generate a light curve of this object, select **Analysis > Histogram plot** (under **Server-side Analysis**). Then, enter **time** in the first field, **1** in the second field, and check the box for bin width, as shown in Figure 4. Select **Run**.

Note that in the resulting light curve, the large gaps of zero counts are a result of data missing from the file, not regions where no radiation was detected from the object.

67. [3 pts] Zoom in on the set of five consecutive bursts around time **181855000**. Estimate the following quantities, in seconds.

- (a) Time between these bursts.
- (b) Duration of each burst.
- (c) Time of the ascending branch (the amount of time taken for the burst to jump from low to high brightness).

68. [2 pts] How is the behavior observed here different from a normal x-ray binary?

69. [2 pts] A recurrent nova is another type of system with recurring bursts. From the above measurements, how can we rule out this object being a recurrent nova?

X-Ray Spectroscopy

The energy spectrum of this object can be modelled as a combination of a thermal blackbody curve and an inverse compton scattering curve, shown in Figure 5.

70. [2 pts] What is inverse compton scattering?

71. [3 pts] When an outburst occurs around this accreting neutron star, why do we see blackbody and inverse compton parts of the spectrum?

Now, generate an energy spectrum for an object. For this dataset, the x -axis is given as a PI channel. This can be converted to energy using

$$E = (\text{PI} \times 0.04 \text{ keV}) + 1.60 \text{ keV}. \quad (2)$$

72. [2 pts] Using Figure 5 as a rough guide, estimate the energy corresponding to the peak blackbody and peak inverse compton distributions, in keV.

Enter column name	time
Enter [min:max:]bins	1
Use bin width instead of number of bins?	☒
Normalize by bin size?	☐

Figure 4: Options to use for generating the object's light curve.

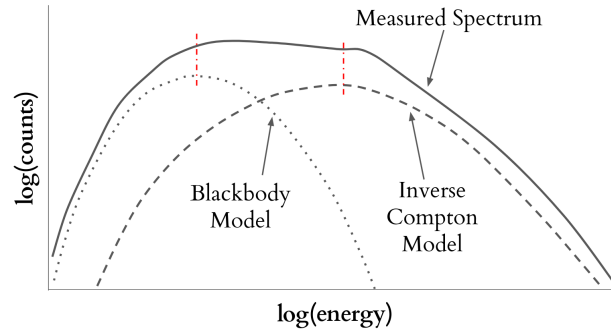


Figure 5: Spectral model for this object.

73. [2 pts] From this, estimate the temperature of the outbursts.
74. [4 pts] We can roughly estimate the energy of a photon that was scattered from inverse compton scattering using

$$E_s = \frac{E_0}{1 - v^2/c^2}, \quad (3)$$

where E_0 is the initial energy of the photon, E_s is the energy of the scattered photon, v is the velocity of the scattering particle, and c is the speed of light.

From this, estimate the speed of the scattering particles, as a fraction of the speed of light.